

Air Pollution Awareness and Environmental Analytics Dashboard

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<https://github.com/subham-cse/Air-Pollution-Awareness-Dashboard>

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Abstract—Urbanization, industrial emissions, and vehicular density have exacerbated ambient air pollution globally, creating severe public health and ecological hazards. Despite the availability of continuous ambient air quality monitoring data, raw sensory metrics often fail to communicate imminent health risks effectively to citizens and environmental analysts. This project synopsis presents the design and implementation of the Air Pollution Awareness Dashboard, an interactive, data-driven analytical web application designed to process, analyze, and visualize ambient air pollutant concentrations ($PM_{2.5}$, PM_{10} , NO_2 , SO_2 , CO , O_3), calculate the National Air Quality Index (AQI), identify spatio-temporal emission trends, and deliver advisory alerts for public health mitigation. Built upon a modern Python stack integrating Streamlit, Pandas, and Plotly, the proposed platform serves as an accessible environmental surveillance and civic awareness tool.

Keywords—Air Pollution Analytics, Air Quality Index (AQI), Environmental Monitoring, Data Visualization, Streamlit, Public Health Awareness, Spatio-Temporal Analysis.

I. INTRODUCTION

Ambient air pollution has emerged as one of the leading global environmental risk factors contributing to cardiopulmonary morbidity and premature mortality. In developing nations and rapidly urbanizing corridors, atmospheric concentrations of hazardous criteria pollutants frequently exceed standard safety thresholds set by the World Health Organization (WHO) and the Central Pollution Control Board (CPCB).

While meteorological stations and regulatory bodies generate continuous time-series streams of pollution metrics, these datasets remain largely inaccessible, complex, and uninterpretable for non-specialist citizens and policymakers. To bridge this information gap, this project proposes the **Air Pollution Awareness Dashboard**—an interactive web-based visualization framework engineered to ingest raw air quality measurements, compute standardized sub-indices, perform comparative and temporal trend analyses, and render intuitive visual indicators and protective health advisories.

II. PROBLEM STATEMENT & OBJECTIVES

Traditional air quality dissemination systems exhibit several key limitations:

1. **High Cognitive Barrier:** Raw numeric pollutant values ($\mu g/m^3$) do not directly convey qualitative health severity to the public.
2. **Lack of Interactive Exploration:** Static reports prevent users from querying region-specific, seasonal, or pollutant-specific historical distributions.
3. **Absence of Contextual Advisories:** Standard feeds of ten state AQI values without providing actionable precau-

tions for vulnerable groups (children, elderly, asthmatics).

Primary Objectives:

- Develop an end-to-end data processing pipeline for multi-pollutant environmental datasets.
- Compute the unified Air Quality Index (AQI) based on standard breakpoint interpolation formulas.
- Provide real-time interactive geospatial, time-series, and correlation analytics using Plotly and Streamlit.
- Generate category-specific health recommendations and civic awareness insights.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The application follows a modular four-tier architecture consisting of Data Ingestion, Processing & AQI Computation, Analytical Visualization, and the User Presentation Layer.

A. Data Ingestion and Preprocessing

The pipeline ingests multi-station atmospheric datasets containing readings for six primary criteria pollutants:

- Fine Particulate Matter ($PM_{2.5}$) and Coarse Particulate Matter (PM_{10})
- Nitrogen Dioxide (NO_2) and Sulfur Dioxide (SO_2)
- Carbon Monoxide (CO) and Ground-level Ozone (O_3)

The data cleaning module handles missing values via forward-filling and time-weighted linear interpolation, detects instrument anomalies, and formats timestamps into granular temporal features (Hour, Day, Month, Season).

B. AQI Mathematical Formulation

The unified AQI is computed using the standard segmented linear interpolation formula:

$$I_p = \frac{I_{hi} - I_{lo}}{BP_{hi} - BP_{lo}} \cdot (C_p - BP_{lo}) + I_{lo} \quad (1)$$

where C_p is the truncated concentration of pollutant p , $[BP_{lo}, BP_{hi}]$ represents the breakpoint concentration interval containing C_p , and $[I_{lo}, I_{hi}]$ is the corresponding AQI sub-index range.

The overall Air Quality Index of a location is determined by the maximum dominant sub-index:

$$AQI = \max(I_{PM_{2.5}}, I_{PM_{10}}, I_{NO_2}, I_{SO_2}, I_{CO}, I_{O_3}) \quad (2)$$

Table 1: AQI Categorization, Range & Health Advisories

AQI Range	Category	Associated Health Advisory
0 – 50	Good	Minimal health impact; safe for outdoor activity.
51 – 100	Satisfactory	Minor breathing discomfort to sensitive people.
101 – 200	Moderate	Discomfort to people with lung/heart disease.
201 – 300	Poor	Breathing discomfort on prolonged exposure.
301 – 400	Very Poor	Respiratory illness on prolonged exposure.
401 – 500	Severe	Severe impact on healthy individuals.

C. Interactive Analytics & Visualization Engine

The dashboard utilizes Plotly for high-fidelity responsive charts:

- **Temporal Trend Lines:** Moving average and multi-scale temporal tracking of pollutant spikes across seasons.
- **Pollutant Breakdown Gauges:** Instantaneous health gauges with color-coded severity indicators.
- **Correlation Matrix Heatmaps:** Statistical dependency analysis between meteorological variables and pollutant dispersion.
- **Comparative Regional Analysis:** Multi-city bar charts and distribution box plots to identify pollution hotspots.

IV. HARDWARE AND SOFTWARE REQUIREMENTS

A. Software Specifications

- **Programming Language:** Python 3.8+
- **Web Framework:** Streamlit
- **Data Manipulation:** Pandas, NumPy
- **Visual Analytics:** Plotly Express, Plotly Graph Objects, Matplotlib, Seaborn
- **Version Control & Repository:** Git, GitHub

B. Hardware Specifications

- **Processor:** Modern Dual-Core / Quad-Core CPU (2.0 GHz or higher)

- **System Memory (RAM):** 4 GB minimum (8 GB recommended)
- **Storage:** 500 MB for repository environment and cached data
- **Display:** Standard web browser (Chrome, Firefox, Edge)

V. EXPERIMENTAL WORKFLOW

1. **Input Layer:** User selects location, pollutant type, and date range filters via the Streamlit sidebar.
2. **Transformation Layer:** Preprocessing module filters, imputes, and aggregates records dynamically.
3. **Analytical Engine:** Mathematical functions calculate dominant pollutant indices and statistical summaries (Mean, Peak, Exceedance count).
4. **Presentation Layer:** The UI updates dynamically, rendering interactive charts, KPIs, AQI dials, and actionable health guidelines.

VI. LIMITATIONS & FUTURE SCOPE

- **Data Latency:** Offline historical datasets require periodic manual synchronizations; real-time sensor API integration is planned.
- **Predictive Analytics:** Future iterations will implement machine learning regression algorithms (e.g., Random Forest, XGBoost, LSTM) to deliver 24-hour predictive AQI forecasts.
- **IoT Integration:** Extension into physical low-cost sensor nodes (ESP32/PMS5003) for localized indoor and outdoor monitoring.

VII. CONCLUSION

The **Air Pollution Awareness Dashboard** provides an intuitive, robust, and interactive platform for environmental data exploration. By converting complex multi-pollutant time-series data into accessible visual metrics and health advisories, the tool democratizes environmental informatics and empowers both individuals and administrative bodies to make informed, health-conscious decisions.

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